

NEXUS-1 — Reactor Physics Reference

The Models Behind the Neutronics and Kinetics Tabs

Equations, constants and numerical methods as implemented in the simulation

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0 Scope and honest framing

This document specifies the physics and numerics actually computed by two screens of the NEXUS-1 console. It is written to be read alongside the source: every equation and constant below is transcribed from the implementation, not idealised.

Two tabs, two different models — read this first. The console deliberately contains two layers of reactor physics, and they are *not* the same code:

- **Reactor Kinetics tab (Part A)** runs a genuine point-kinetics solver: six delayed-neutron groups, Doppler / moderator / xenon feedback, a two-node thermal model, and a stable operator-splitting integrator. Its *structure* is the real physics of reactor dynamics.
- **Neutronics tab (Part B)** is driven by the lightweight per-unit *fleet* model plus display heuristics. Its excore-detector readings, reactivity-balance rows and axial-flux shape are illustrative mappings chosen to behave plausibly — they are *not* outputs of the point-kinetics solver.

In both parts, numerical **constants** (generation time, feedback coefficients, thermal gains, xenon worth, detector scalings) are representative textbook-order values, *not* tuned to any specific reactor. This is a demonstration and educational tool, not a licensed analysis code, and it is not connected to any real plant.

Notation: relative neutron population / relative power n (with $n = 1$ at full power), reactivity in pcm (1 pcm = $10^{-5} \Delta k/k$), temperatures in °C, time in seconds unless stated.

1 Part A — Reactor Kinetics tab (rigorous model)

1.1 A.1 Point-kinetics equations

The core of this tab integrates the point-kinetics equations with six delayed-neutron precursor groups (U-235 thermal data):

$$\frac{dn}{dt} = \frac{\rho - \beta}{\Lambda} n + \sum_{i=1}^6 \lambda_i C_i, \quad (1)$$

$$\frac{dC_i}{dt} = \frac{\beta_i}{\Lambda} n - \lambda_i C_i, \quad i = 1 \dots 6, \quad (2)$$

where n is the relative neutron population, C_i the precursor concentration of group i , ρ the net reactivity ($\Delta k/k$), $\beta = \sum_i \beta_i$ the total delayed fraction, Λ the prompt-neutron generation time, and λ_i the precursor decay constants. At steady state ($dC_i/dt = 0$) the precursors satisfy $C_i = \beta_i n / (\Lambda \lambda_i)$, which is exactly how the equilibrium state is seeded.

Group	i	β_i	λ_i (s ⁻¹)
	1	0.000 215	0.0124
	2	0.001 424	0.0305
	3	0.001 274	0.1110
	4	0.002 568	0.3010
	5	0.000 748	1.1400
	6	0.000 273	3.0100
$\beta = \sum_i \beta_i \approx 0.006502 \quad \bullet \quad \Lambda = 2 \times 10^{-5} \text{ s}$			

Table 1: *

Delayed-neutron data and prompt generation time (BETA, LAM, GEN).

1.2 A.2 Reactivity decomposition

The instantaneous reactivity (in pcm) is the sum of the rod contribution and three feedbacks:

$$\rho_{\text{pcm}} = \underbrace{\rho_{\text{rod}} + \rho_{\text{scram}}}_{\text{control}} + \underbrace{a_D (T_F - T_{F0})}_{\text{Doppler}} + \underbrace{a_M (T_C - T_{C0})}_{\text{moderator}} + \underbrace{-K_{Xe} (Xe - Xe_{\text{eq}})}_{\text{xenon-135}} \quad (3)$$

with the SCRAM term $\rho_{\text{scram}} = -8000$ pcm while a trip is active (otherwise 0). The absolute reactivity used in the kinetics equations is $\rho = \rho_{\text{pcm}} \times 10^{-5}$.

Coefficient	Value	Meaning
a_D	-2.5 pcm/°C	Doppler (fuel-temperature) feedback
a_M	-15.0 pcm/°C	moderator (coolant-temperature) feedback
T_{F0}, T_{C0}	600, 305 °C	reference fuel / coolant temperatures
K_{Xe}	2700/ Xe_{eq}	scales xenon to ≈ 2700 pcm equilibrium worth

Both feedback coefficients are negative, giving the stabilising self-regulation characteristic of a well-behaved thermal reactor.

1.3 A.3 Two-node thermal model

Fuel and coolant temperatures follow a lumped two-node model driven by relative power $P \equiv n$:

$$C_F \frac{dT_F}{dt} = P K_P - K_{FC} (T_F - T_C), \quad (4)$$

$$C_C \frac{dT_C}{dt} = K_{FC} (T_F - T_C) - K_{CO} (T_C - T_{\text{in}}). \quad (5)$$

The fuel node is heated in proportion to power and loses heat to the coolant; the coolant node receives that heat and rejects it to the inlet sink T_{in} .

K_P	K_{FC}	K_{CO}	C_F	C_C	T_{in}
59	0.2	3.93	2	4	290 °C

1.4 A.4 Xenon–iodine poisoning

Xenon-135 is governed by the coupled iodine/xenon balance, with iodine produced directly from fission and decaying into xenon, and xenon additionally burned out by neutron capture ($\sigma\phi$ term):

$$\frac{dI}{dt} = \gamma_I S_f P - \lambda_I I, \quad (6)$$

$$\frac{dXe}{dt} = \gamma_{Xe} S_f P + \lambda_I I - \lambda_{Xe} Xe - \sigma\phi P Xe. \quad (7)$$

The equilibrium values used to seed the state and to reference the xenon feedback are

$$I_{\text{eq}} = \frac{\gamma_I S_f}{\lambda_I}, \quad Xe_{\text{eq}} = \frac{\gamma_{Xe} S_f + \gamma_I S_f}{\lambda_{Xe} + \sigma \phi}. \quad (8)$$

γ_I	0.0639	λ_I	$2.87 \times 10^{-5} \text{ s}^{-1}$
γ_{Xe}	0.00228	λ_{Xe}	$2.09 \times 10^{-5} \text{ s}^{-1}$
$\sigma \phi$	0.7×10^{-4}	S_f	1 (fission source, full power)

1.5 A.5 Numerical integration (operator splitting)

The point-kinetics equations are stiff: $\Lambda \sim 10^{-5} \text{ s}$ sits next to precursor and thermal timescales of seconds to hours. A naive explicit scheme would be unstable, so the integrator splits the update by timescale.

(1) Precursors — exact update. Holding n constant over a step dt , each precursor group has an analytic solution:

$$C_i \leftarrow C_i e^{-\lambda_i dt} + \frac{\beta_i n}{\Lambda \lambda_i} (1 - e^{-\lambda_i dt}), \quad S_{\text{src}} = \sum_{i=1}^6 \lambda_i C_i. \quad (9)$$

(2) Neutron population — implicit Euler. The fast neutron equation is advanced implicitly for unconditional stability, using a reactivity clamped to $\rho_U = \min(\rho, 0.95 \beta)$ so the denominator stays well-conditioned near prompt criticality:

$$n \leftarrow \frac{n + dt S_{\text{src}}}{1 - dt (\rho_U - \beta) / \Lambda}. \quad (10)$$

(3) Thermal and xenon — explicit Euler. The slow nodes (T_F, T_C, I, Xe) are advanced with simple forward Euler, which is accurate and stable at their long timescales.

Substepping. Each animation frame advances $dt_{\text{real}} = 0.05 \times \text{timeScale}$ simulated seconds, subdivided into up to 500 inner steps of $\approx 0.02 \text{ s}$ so the integrator stays accurate even at high time-acceleration (e.g. 600 $\times$ for watching the xenon transient over hours).

1.6 A.6 Derived and displayed quantities

- **Relative power:** $P = n$ (neutron population equals relative power in this model).
- **Reactor period** $\tau = n/\dot{n}$, reported as ∞ when $|\dot{n}|$ is negligible; here $\dot{n} = \frac{\rho - \beta}{\Lambda} n + S_{\text{src}}$.
- **Reactivity in dollars:** $\$ = \rho_{\text{pcm}} / (\beta \times 10^5) \approx \rho_{\text{pcm}} / 650$, so \$1 corresponds to one full β of reactivity.
- **Prompt-critical flag:** raised when $\rho \geq \beta$, the threshold above which the prompt neutrons alone sustain the chain reaction.

1.7 A.7 What the model reproduces

Because the *structure* is the real physics, the tab reproduces the correct qualitative phenomena even with representative constants: the **prompt jump** after a positive reactivity step, the resulting **reactor period**, **power self-regulation** through negative temperature feedback (a step settles to a new power rather than diverging), the **delayed-neutron tail** that prevents power from collapsing instantly on SCRAM, and the slow **post-shutdown xenon transient** building over hours.

2 Part B — Neutronics tab (illustrative model)

The Neutronics tab does *not* call the Part-A solver. Its numbers come from the per-unit fleet update (`stepUnit`) plus monotonic display mappings chosen to look like real instrumentation. Treat the formulas below as *presentation logic*, not first-principles neutronics.

2.1 B.1 Fleet unit dynamics (the driver)

For the selected unit, rod motion relaxes toward a target and flux follows a first-order response to a rod-derived target, with small noise ε :

$$\text{rodInsert} \leftarrow \text{rodInsert} + (\text{target} - \text{rodInsert}) \cdot k_{\text{ease}}, \quad (11)$$

$$\phi_{\text{target}} = \text{clamp}(100 - (\text{rodInsert} - 34) \cdot 2.2, 0, 108), \quad (12)$$

$$\phi \leftarrow \phi + (\phi_{\text{target}} - \phi) \cdot 0.18 + \varepsilon, \quad \phi \in [0, 110], \quad (13)$$

$$\rho_{\text{disp}} = \text{clamp}((\phi_{\text{target}} - \phi) \cdot 9 + \varepsilon, -2600, 260) \text{ pcm}. \quad (14)$$

Here ϕ is relative power/flux in %, and ρ_{disp} is a proxy reactivity that is positive while flux is still climbing toward its target and settles to zero in steady state. The “34” offset is the nominal regulating-bank insertion at full power.

2.2 B.2 Excore detector mappings

The three instrument ranges are simple monotonic functions of flux and the proxy reactivity, scaled to realistic display units:

$$\text{Source range} = 18 + 0.4 \phi \quad [\text{cps}], \quad (15)$$

$$\text{Intermediate range} = \frac{\phi}{100} \cdot 4.2 \times 10^{-4} \quad [\text{A}], \quad (16)$$

$$\text{Startup rate (SUR)} = \rho_{\text{disp}}/55 \quad [\text{DPM}]. \quad (17)$$

These give the correct *behaviour* an operator expects — count rate and detector current rise with power, startup rate tracks the sign and size of reactivity — without modelling detector physics.

2.3 B.3 Reactivity-balance panel

The panel sums live and static contributions:

Row	Value	Type
Control rods	$-18 \cdot \text{rodInsert}$ pcm	live (tracks rod position)
Xenon-135	$-46 \cdot \text{xenon}$ pcm	live (tracks fleet xenon state)
Boron (soluble)	-1180 pcm	static placeholder
Moderator temp. coeff.	-28 pcm/°C	static label
Doppler (fuel temp.)	-2.4 pcm/°C	static label
Net reactivity	ρ_{disp}	live

The net row is the live proxy reactivity from B.1; the rod and xenon rows move with the simulation, while boron and the two coefficient rows are fixed illustrative values typical of a PWR at power.

2.4 B.4 Axial flux profile

The axial shape is rendered over $n = 12$ bins from top ($f = 0$) to bottom ($f = 1$), with $f = i/(n-1)$ and rod insertion fraction $\text{ins} = \text{rodInsert}/100$:

$$\text{shape}(f) = \sin(\pi (0.08 + 0.84 f)), \quad (18)$$

$$\text{suppress}(f) = \text{clamp}(1 - (\text{ins} - f) \cdot 1.4, 0.05, 1), \quad (19)$$

$$\phi(f) = \text{clamp}(\text{shape}(f) \cdot \text{suppress}(f), 0, 1) \cdot \frac{\phi}{100}. \quad (20)$$

The cosine-like base shape gives the familiar peaked axial distribution; the suppression term damps the upper core as rods insert from the top, shifting the peak downward — a visual stand-in for **axial offset**. Bins above the rod tip ($f < \text{ins}$) are coloured to mark the rodded region.

2.5 B.5 Period on this tab

The neutronics/core period indicator uses a heuristic inverse of the proxy reactivity (distinct from the rigorous $\tau = n/\dot{n}$ of Part A):

$$\tau = \text{clamp}(26000/|\rho_{\text{disp}}|, 2, 999) \text{ s}, \quad \tau = \infty \text{ when } |\rho_{\text{disp}}| < 40. \quad (21)$$

3 Part C — Relationship, limitations and roadmap

Why two models. The fleet model (Part B) is intentionally cheap: it runs for every unit, every tick, to drive dozens of plant screens. The point-kinetics solver (Part A) is heavier and runs for the one unit on display in the Kinetics tab. Keeping them separate was a performance choice during Phase 0.

Known limitation. The cost of that split is that the Neutronics and Kinetics tabs can disagree about “what the reactor is doing,” because they compute it differently. For a digital-twin project this is the one inconsistency worth retiring: a single point-kinetics engine, parameterised per unit and instantiated across the fleet, would make every tab a view of one source of truth. That unification is a tracked roadmap item.

Validation. The Part-A solver can be checked against analytic cases — asymptotic period for a given reactivity step, the prompt-jump ratio, and xenon equilibrium / post-shutdown peak timing — which is the natural next addition to this document.

Disclaimer. NEXUS-1 is a personal demonstration and educational project applying software-architecture, simulation and visualisation skills to a studied domain. It is not a claim of nuclear-engineering expertise, not a licensed analysis tool, and not connected to any real reactor. Physics constants are representative textbook-order values. Build NX1-094C79E0A366D34F • © 2026 Grigorios Agathangelidis.